

Background magnetic field stabilizes QCD string against breaking

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The confinement of quarks in hadrons occurs due to formation of QCD string. At large separation between the quarks the QCD string breaks into pieces due to light quark-antiquark pair creation. We argue that there exist a critical background magnetic field $eB \simeq 16m_\pi^2$, above which the string breaking is impossible in the transverse directions with respect to the axis of the magnetic field. Thus, at strong enough magnetic field a new, asymmetrically confining phase may form. The effect – which can potentially be tested at LHC/ALICE experiment – leads to abundance of u -quark rich hadrons and to excess of radially excited mesons in the noncentral heavy-ion collisions compared to the central ones.

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Motivation. Noncentral heavy-ion collisions create intense magnetic fields with the magnitude of the order of the QCD scale. According to [1] the strength of the emerging magnetic field B may reach

$$eB_{\text{RHIC}}^{\text{max}} \sim m_\pi^2, \quad eB_{\text{LHC}}^{\text{max}} \sim 15 m_\pi^2 \quad (1)$$

at the Relativistic Heavy Ion Collider (RHIC) and at the Large Hadron Collider (LHC). Here $e = |e|$ is the absolute value of the electron charge.

There are various potentially observable QCD effects associated with the presence of the strong magnetic field background. One can mention a CP-odd generation of an electric current of quarks along the axis of the magnetic field (“the chiral magnetic effect”) [2] and enhancement of the chiral symmetry breaking (“the magnetic catalysis”) [3]. The latter is related to the fact that the background magnetic field makes the chiral condensate larger [4]. Acting through the chiral condensate, the magnetic field also shifts and strengthens the chiral transition [5].

Recently, observation of certain signatures of the chiral magnetic effect was reported by the STAR collaboration at the RHIC experimental facility [6]. Some effects were also found in numerical simulations of lattice QCD. There exists numerical evidence in favor of existence of both the chiral magnetic effect [7] and the enhancement of the chiral symmetry breaking [8]. In addition, a chiral magnetization of the QCD vacuum – discussed first in Ref. [9] – was calculated numerically [10]. The lattice simulations have also revealed that due to CP-odd structure of the QCD vacuum the quark’s magnetic dipole moment in a strong magnetic field gets a large CP-odd piece, the electric dipole moment [11].

In addition to the chiral properties, the background magnetic field should also be important for confining features of QCD despite the photons are not interacting with the gluons directly. A strong enough magnetic field affects the dynamics of the gluons through the in-

fluence on the quarks, because the quarks are coupled to the both gauge fields [12]. And, indeed, thermodynamic arguments of Ref. [13] suggest that the background magnetic field should shift and weaken the confinement-deconfinement phase transition in the QCD vacuum. The confining and deconfining regions at zero temperature are separated by a smooth crossover which is located at [13]

$$eB_{\text{cross}}[T=0] \sim (700 \text{ MeV})^2 \sim 25 m_\pi^2. \quad (2)$$

Due to the crossover nature of the transition, the difference between confining and deconfining regions is somewhat obscure. Therefore the confining interaction between the quarks may also be visible at the magnetic fields that are stronger than the crossover scale (2).

Although in our paper we discuss QCD vacuum effects of the strong magnetic fields below the crossover scale (2), it is interesting to mention that at stronger magnetic fields, $eB \gtrsim 150 m_\pi^2$ an exotic condensation of the $u\bar{u}$ pairs should occur [14]. Effects induced by magnetic field at finite quark density are also dramatic. For example, at $eB \gtrsim 3m_\pi^2$ a stack of parallel π^0 domain walls should become more favorable (from an energy-related point of view) than nuclear matter at the same density [15].

Confinement and string breaking at $B = 0$. The quark confinement exists because the chromoelectric flux (in, say, a test quark-antiquark pair) is squeezed into a stringlike structure, the QCD string, due to nonperturbative vacuum effects. The QCD string has a nonzero tension σ , and therefore the energy of the long enough string is proportional to its length R ,

$$V_{\text{str}}(R) = \sigma R, \quad (3)$$

In a quarkless QCD (i.e., in Yang-Mills theory) the energy of the QCD string (3) rises infinitely with the separation between test quark and antiquark. Inevitably, the unboundedness of the potential leads to the quark confinement. However, in the real QCD the string breaks